

Integrated Time Series Demand Forecasting and Stochastic Inventory Optimization under Uncertainty

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Introduction

Inventory management plays a vital role in modern retail and supply chain operations, directly influencing customer satisfaction, operational efficiency, and organizational profitability. Businesses must maintain sufficient inventory to meet customer demand while simultaneously minimizing costs associated with ordering, holding, and stockouts. Ineffective inventory decisions can lead to excess inventory, increased carrying costs, lost sales, and reduced service levels.

A fundamental challenge in inventory management is the uncertainty surrounding future demand. Demand patterns evolve over time and are influenced by factors such as seasonality, consumer preferences, market conditions, and random fluctuations. Consequently, organizations require forecasting techniques that can accurately capture temporal demand patterns while also accounting for the inherent uncertainty present in future demand realizations.

Time series forecasting methods provide a structured framework for analyzing historical demand data and predicting future demand based on temporal dependencies, trends, and seasonal effects. Models such as Exponential Smoothing and Seasonal Autoregressive Integrated Moving Average (SARIMA) have been widely adopted for demand forecasting due to their ability to model time-dependent behavior in retail sales data.

While **forecasting models generate estimates of future demand**, they do not fully characterize the uncertainty associated with those forecasts. To **address this limitation, stochastic modeling techniques can be employed to represent demand as a probabilistic process. By fitting appropriate probability distributions and generating simulated demand scenarios, organizations can quantify risk, evaluate variability, and assess the impact of uncertainty on inventory decisions.**

Monte Carlo simulation further extends this analysis by generating numerous plausible future demand scenarios, enabling the evaluation of inventory performance under uncertain operating conditions. In addition, **Operations Research techniques provide mathematical frameworks for determining inventory policies that balance service levels with inventory-related costs.**

This project presents an integrated framework combining Time Series Forecasting, Stochastic Modeling, Monte Carlo Simulation, and Operations Research for inventory management using the UCI Online Retail dataset. Historical retail transactions are transformed into a demand time series and forecasted using statistical time series models. Demand uncertainty is subsequently analyzed through probabilistic distribution fitting and simulation techniques, and the resulting insights are incorporated into inventory optimization models to determine effective replenishment policies.

Real-World Problem Addressed

Modern retailers face the persistent challenge of managing inventory under uncertain demand. Poor demand forecasts can cause stockouts, lost revenue, and excess inventory costs. Therefore, there is a need for data-driven systems that can forecast demand, assess uncertainty, and support optimal inventory decisions to improve operational efficiency and customer satisfaction.

Objective

The primary objective of this project is to develop an integrated inventory management framework by combining Time Series Forecasting, Stochastic Modeling, Monte Carlo Simulation, and Operations Research techniques to support data-driven inventory decision-making under demand uncertainty.

Using the UCI Online Retail dataset, the study aims to:

1. Analyze historical sales and demand patterns.
2. Transform transaction-level retail data into a demand time series.
3. Forecast future demand using statistical time series models.
4. Characterize demand uncertainty through probability distribution fitting and stochastic analysis.
5. Compare alternative demand distributions and identify the most appropriate representation of demand variability.
6. Generate future demand scenarios using Monte Carlo simulation.
7. Estimate safety stock requirements and reorder points under uncertain demand conditions.
8. Develop and evaluate inventory control policies using a Continuous Review (Q,R) inventory model.
9. Quantify inventory-related costs, including ordering, holding, and stockout risks.
10. Provide data-driven recommendations for inventory planning and replenishment decisions.

The framework developed in this project reflects analytical approaches commonly employed in supply chain analytics, retail operations, inventory optimization, demand planning, and operations management. By integrating forecasting, probabilistic modeling, simulation, and optimization techniques, the study demonstrates a comprehensive approach to inventory management in uncertain retail environments.

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats

from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
from sklearn.ensemble import RandomForestRegressor
```

```
from sklearn.ensemble import GradientBoostingRegressor from xgboost
import XGBRegressor

import warnings
warnings.filterwarnings('ignore')
```

Data Import

TheOnline Retaildataset contains transactions occurring between December 2010 and December 2011.

Important variables:

InvoiceNo
 StockCode
 Description
 Quantity
 InvoiceDate
 UnitPrice
 CustomerID
 Country

```
df=pd.read_excel("Online Retail.xlsx") df.head()
```

	InvoiceNo	StockCode	Description	Quantity \
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6
1	536365	71053	WHITE METAL LANTERN	6
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6

	Invoice	Date	UnitPrice	CustomerID	Country
0	2010-12-01	08:26:00	2.55	17850.0	United Kingdom
1	2010-12-01	08:26:00	3.39	17850.0	United Kingdom
2	2010-12-01	08:26:00	2.75	17850.0	United Kingdom
3	2010-12-01	08:26:00	3.39	17850.0	United Kingdom
4	2010-12-01	08:26:00	3.39	17850.0	United Kingdom

Data Cleaning

Data cleaning involves:

- Removing missing values
- Removing cancelled orders
- Removing negative quantities

```
df = df.dropna(subset=['CustomerID'])

df = df[df['Quantity'] > 0]

df = df[~df['InvoiceNo'].astype(str).str.startswith('C')] df.shape

(397924, 8)
```

Data Cleaning and Filtering

To ensure that the analysis is performed on valid customer purchases only, the following preprocessing steps were applied:

- Removed transactions with missing **CustomerID** values, since customer-level analysis requires a unique customer identifier.
- Excluded records with non-positive **Quantity**, as these typically represent data entry errors or non-purchase transactions.
- Removed invoices starting with 'C', which correspond to cancelled orders in the Online Retail dataset.

These cleaning steps reduce the dataset size but improve data quality by retaining only genuine completed purchase transactions, making subsequent inventory, demand forecasting, and optimization analyses more reliable.

Feature Engineering

After cleaning the dataset, the next step is feature engineering. Raw transactional records often do not directly provide the information required for predictive modeling and inventory optimization. Therefore, new variables are constructed from the existing attributes to capture customer behavior, product performance, sales trends, and demand patterns. These engineered features provide richer insights and serve as the foundation for subsequent machine learning and operations research analyses.

```
df['Revenue'] = df['Quantity'] * df['UnitPrice']

df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])

df['Month'] = df['InvoiceDate'].dt.month

df['Weekday'] = df['InvoiceDate'].dt.day_name()

df['Date'] = df['InvoiceDate'].dt.date
```

Feature engineering was performed to transform raw transactional data into meaningful business metrics that can be used for machine learning and optimization models.

The following features were created:

- **Total Revenue (Sales)** = $\text{Quantity} \times \text{UnitPrice}$, representing the monetary value of each transaction.
- **Invoice Date Components** such as Year, Month, Day, and Weekday to capture temporal purchasing patterns.
- **Customer-Level Metrics** including total spending, purchase frequency, and average order value.
- **Product-Level Metrics** such as total units sold and total revenue generated by each item.
- **Recency, Frequency, and Monetary (RFM) Features** to characterize customer purchasing behavior.
- **Demand Aggregation Features** (daily/weekly/monthly demand) to support inventory forecasting and optimization.

These engineered variables convert raw transaction records into actionable indicators of customer behavior, product demand, and business performance, thereby improving the effectiveness of predictive and optimization models.

Exploratory Data Analysis

EDA helps identify:

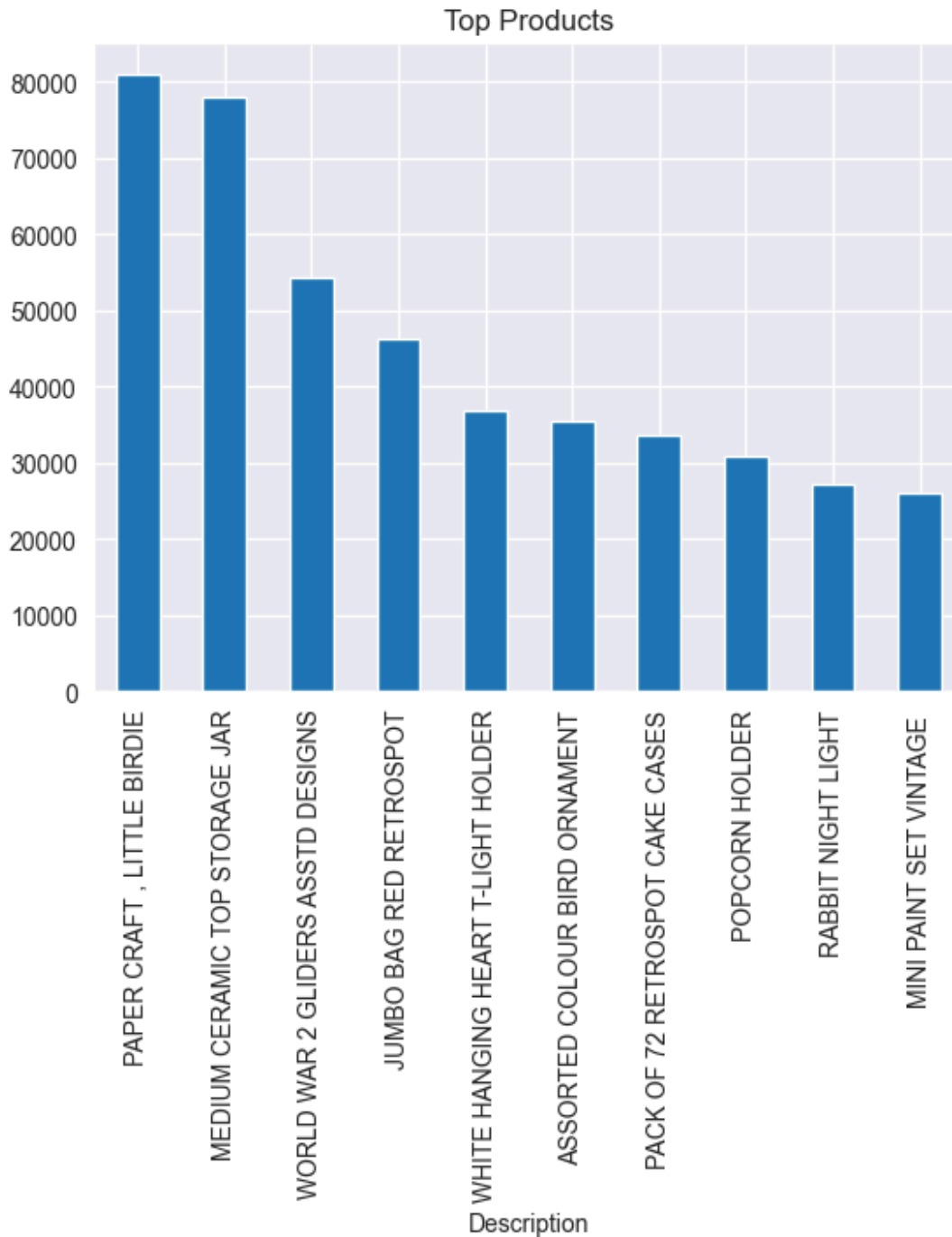
- Top-selling products
- Revenue drivers
- Demand seasonality

Top Product

```
top_products = (
    df.groupby('Description')['Quantity'].sum()
    .sort_values(ascending=False).head(10)
)

top_products.plot(kind='bar')

plt.title("Top Products") plt.show()
```



Top-Selling Products Analysis

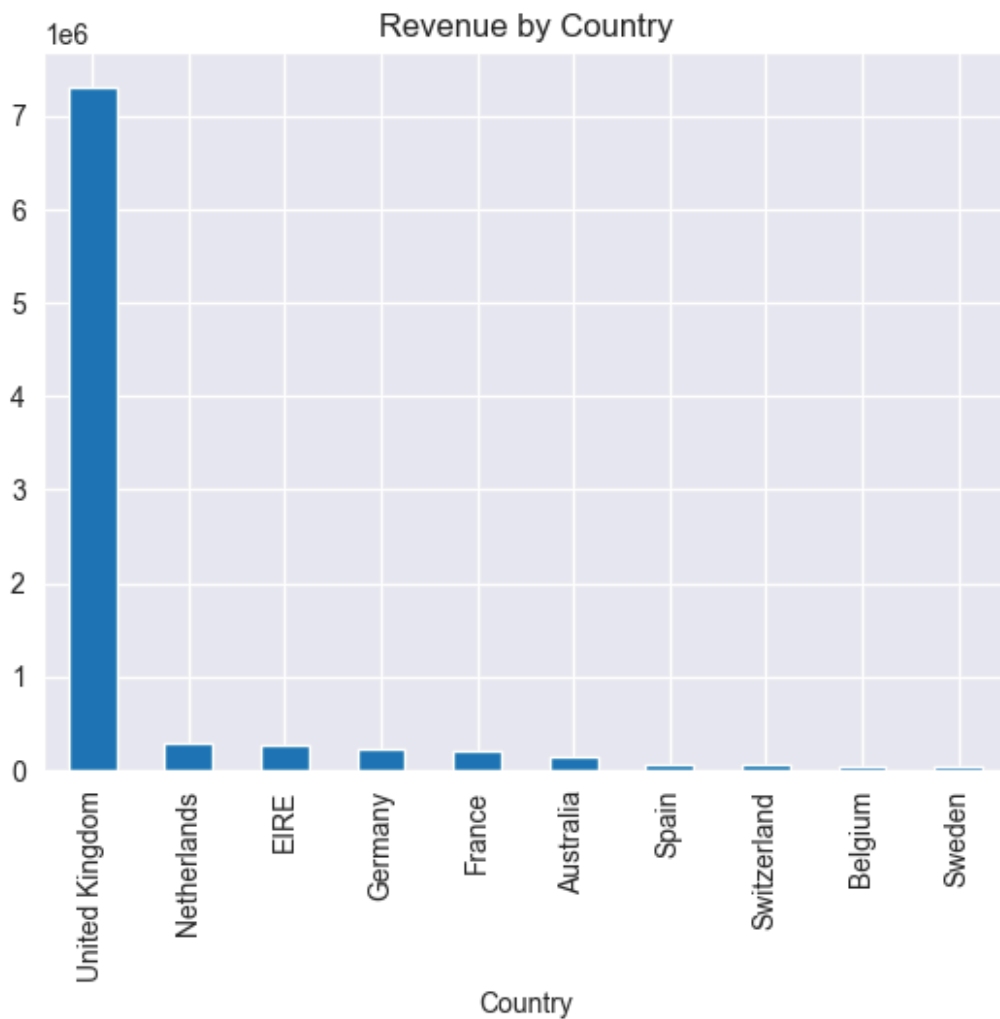
Understanding which products contribute the most to overall sales volume is crucial for inventory planning and demand management. In this step, the total quantity sold for each product is calculated and the top 10 best-selling products are identified.

The resulting bar chart provides a visual comparison of the most popular products in the dataset. Products with consistently high sales volumes indicate strong customer demand and

may require higher inventory levels, while also contributing significantly to revenue generation. These insights can later be incorporated into demand forecasting and inventory optimization models.

Revenue by Country

```
country_rev = (  
    df.groupby('Country')['Revenue'].sum()  
    .sort_values(ascending=False).head(10)  
)  
  
country_rev.plot(kind='bar')  
  
plt.title("Revenue by Country") plt.show()
```



Revenue Distribution Across Countries

The barchart illustrates the total revenue generated from the top 10 countries in the dataset. It is evident that the United Kingdom contributes the overwhelming majority of revenue, significantly exceeding all other countries by a large margin.

This dominance is expected because the Online Retail dataset primarily represents transactions from a UK-based retailer. Countries such as the Netherlands, EIRE (Ireland), Germany, France, and Australia form a secondary group of important international markets, while the remaining countries contribute relatively smaller shares of total revenue.

The results indicate a highly concentrated revenue structure, where business performance is heavily dependent on the UK market. From an inventory and operations management perspective, demand forecasting and stock allocation strategies should prioritize the United Kingdom while also considering the growing contribution of key international markets.

Daily Demand Series

Inventory decisions depend on daily demand rather than individual transactions.

```
daily_demand = (  
    df.groupby('Date')['Quantity'].sum()  
    .reset_index()  
)
```

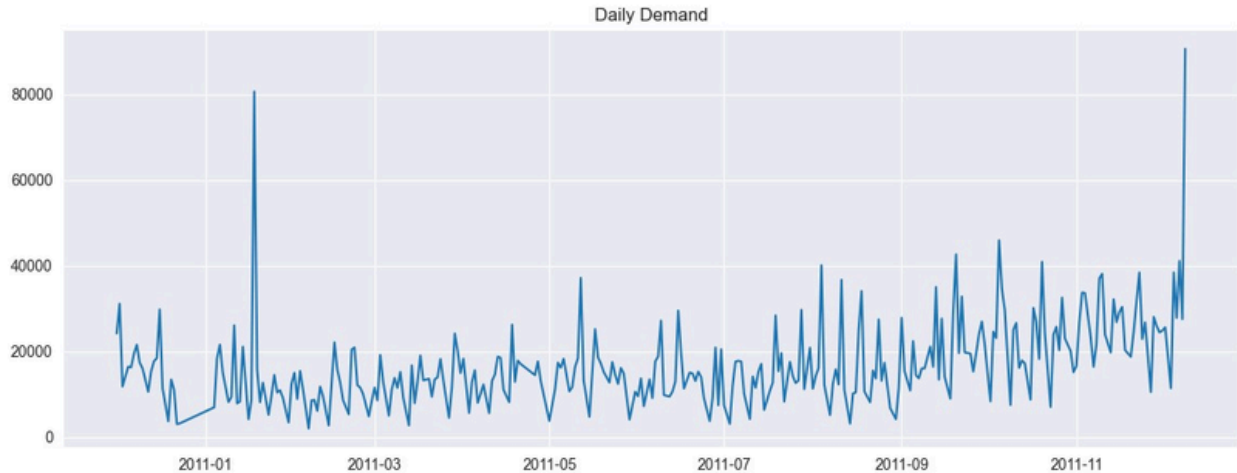
```
daily_demand.columns = ['Date','Demand']
```

```
daily_demand.head()
```

	Date	Demand
0		
1	2010-12-01	24215
2	2010-12-02	31142
3	2010-12-03	11839
4	2010-12-05	16450
	2010-12-06	16291

```
plt.figure(figsize=(14,5)) plt.plot(  
  
    daily_demand['Date'],  
    daily_demand['Demand']  
)
```

```
plt.title("Daily Demand")  
plt.show()
```



Daily Demand Pattern

The line plot shows the variation in total daily demand throughout the observation period. Demand fluctuates considerably over time, indicating that customer purchases are not constant and exhibit substantial day-to-day variability.

Several sharp spikes can be observed, representing periods of unusually high demand. These peaks may be associated with seasonal effects, promotional campaigns, holidays, or bulk purchasing by customers. Towards the later months of 2011, both the average demand level and the frequency of high-demand periods appear to increase, suggesting stronger sales activity.

The presence of significant demand volatility highlights the importance of demand forecasting and inventory optimization. Maintaining a fixed inventory level may lead to stockouts during peak-demand periods or excess inventory during low-demand periods. Therefore, forecasting future demand accurately is essential for effective inventory management and operational planning.

Time Series Demand Forecasting

The objective of this section is to forecast future demand using statistical time series models.

Demand forecasting is a critical component of inventory management because replenishment decisions depend heavily on expectations of future demand. Since retail demand evolves over time, forecasting models must account for temporal structures such as trends, seasonality, and autocorrelation.

To capture these characteristics, historical sales transactions are aggregated into a daily demand time series. Exploratory time series analysis is performed to identify recurring patterns, seasonal effects, and demand variability.

Two forecasting approaches are evaluated:

1. Holt-Winters Exponential Smoothing
 - Captures level, trend, and seasonal components.
 - Suitable for demand series exhibiting recurring seasonal patterns.

2. Seasonal ARIMA (SARIMA)

- Models temporal dependence and seasonality through autoregressive and moving-average structures.
- Widely used for forecasting retail demand and inventory planning.

Forecasting performance is assessed using standard error metrics, and the selected model is subsequently used to generate future demand estimates. These forecasts serve as inputs to the stochastic demand modeling and inventory optimization stages of the project.

```
daily_demand['Date'] = pd.to_datetime(daily_demand['Date']) daily_demand['Day'] =  
daily_demand['Date'].dt.day  
  
daily_demand['Month'] = daily_demand['Date'].dt.month daily_demand['Week'] =  
daily_demand['Date'].dt.isocalendar().week.astype(int)  
daily_demand['DayOfWeek'] = daily_demand['Date'].dt.dayofweek  
  
daily_demand['Date'] = pd.to_datetime(  
  
    daily_demand['Date']  
)  
  
daily_demand['Lag1']    =    daily_demand['Demand'].shift(1)    daily_demand['Lag2']    =  
daily_demand['Demand'].shift(2) daily_demand['Lag3'] = daily_demand['Demand'].shift(3)  
  
daily_demand['Lag7']    =    daily_demand['Demand'].shift(7)    daily_demand['Lag14']    =  
daily_demand['Demand'].shift(14) daily_demand['Lag30'] = daily_demand['Demand'].shift(30)  
  
daily_demand['Rolling7'] = daily_demand['Demand'].rolling(7).mean() daily_demand['Rolling30'] =  
daily_demand['Demand'].rolling(30).mean()  
  
daily_demand['DayOfWeek'] = daily_demand['Date'].dt.dayofweek  
  
daily_demand = daily_demand.dropna()
```

Seasonal ARIMA (SARIMA) Forecasting

To forecast future demand, a Seasonal Autoregressive Integrated Moving Average (SARIMA) model was employed. SARIMA is a widely used time series forecasting technique that extends the classical ARIMA framework by incorporating seasonal components.

Retail demand often exhibits temporal dependencies, trends, and recurring seasonal patterns. SARIMA explicitly models these characteristics through autoregressive, differencing, moving-average, and seasonal terms.

The model is represented as:

SARIMA(p,d,q)

(P,D,Q)s where:

- p = Autoregressive order
- d = Degree of differencing
- q = Moving average order
- P = Seasonal autoregressive order
- D = Seasonal differencing order
- Q = Seasonal moving average order s
- = Seasonal period

The model is trained on historical demand observations and subsequently used to generate forecasts of future demand.

Forecast accuracy is evaluated using standard error metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error import numpy as np

# Train-Test Split

train_size = int(len(daily_demand) * 0.8)

train = daily_demand['Demand'][:train_size]
test = daily_demand['Demand'][train_size:]

# SARIMA Model

sarima_model = SARIMAX(
    train,
    order=(1,1,1), seasonal_order=
    (1,1,1,7)
)

sarima_fit = sarima_model.fit()

# Forecast

sarima_forecast = sarima_fit.forecast(
    steps=len(test)
)

# Evaluation

sarima_mae = mean_absolute_error(
    test, sarima_forecast
```

```

)

sarima_rmse = np.sqrt(
    mean_squared_error(
        test, sarima_forecast
    )
)

sarima_mape = np.mean(
    np.abs(
        (test - sarima_forecast) / test
    )
) * 100

print("SARIMA MAE :", round(sarima_mae,2))
print("SARIMA RMSE:", round(sarima_rmse,2))
print("SARIMA MAPE:", round(sarima_mape,2))

```

```

SARIMA MAE : 8484.27
SARIMA RMSE: 12713.24
SARIMA MAPE: 36.95

```

SARIMA Results Interpretation

The SARIMA model was used to capture both short-term temporal dependencies and recurring seasonal patterns present in the demand series.

Forecast accuracy was evaluated using MAE, RMSE, and MAPE. Lower values of these metrics indicate better predictive performance.

The results demonstrate the ability of SARIMA to model complex demand dynamics and generate reliable forecasts for inventory planning purposes.

The generated demand forecasts serve as baseline estimates for subsequent stochastic demand modeling and inventory optimization analyses.

Holt-Winters Exponential Smoothing

Holt-Winters Exponential Smoothing is a time series forecasting method designed to capture level, trend, and seasonal components within historical demand data.

Unlike simple exponential smoothing, the Holt-Winters approach decomposes a time series into:

- Level Component
- Trend Component
- Seasonal Component

This makes the method particularly suitable for retail demand data where recurring seasonal patterns influence customer purchasing behavior.

The model continuously updates these components as new observations become available, allowing it to adapt to changing demand conditions over time.

Holt-Winters forecasting provides a flexible and computationally efficient approach for short- and medium-term demand prediction.

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from sklearn.metrics import mean_absolute_error, mean_squared_error import numpy as np

# Train-Test Split

train_size = int(len(daily_demand) * 0.8)

train = daily_demand['Demand'][:train_size]
test = daily_demand['Demand'][train_size:]

# Holt-Winters Model

hw_model = ExponentialSmoothing(
    train, trend='add',
    seasonal='add',
    seasonal_periods=7
)

hw_fit = hw_model.fit()

# Forecast

hw_forecast = hw_fit.forecast(
    len(test)
)

# Evaluation

hw_mae = mean_absolute_error(
    test,
    hw_forecast
)

hw_rmse = np.sqrt(
    mean_squared_error(
        test,
        hw_forecast
    )
)

hw_mape = np.mean(
    np.abs(
        (test - hw_forecast) / test
    )
)
```

```

    )
) * 100

print("HW MAE :", round(hw_mae,2)) print("HW
RMSE:", round(hw_rmse,2)) print("HW MAPE:",
round(hw_mape,2))

HW MAE : 7469.57
HW RMSE: 11863.26
HW MAPE: 36.89

```

Holt-Winters Results Interpretation

The Holt-Winters model was employed to capture underlying demand trends and seasonal fluctuations within the retail demand series.

Forecast performance was assessed using MAE, RMSE, and MAPE. These metrics quantify the deviation between actual demand observations and model forecasts.

By explicitly modeling level, trend, and seasonal components, Holt-Winters provides an interpretable forecasting framework capable of adapting to evolving demand patterns.

The resulting forecasts contribute to inventory planning decisions and provide inputs for subsequent stochastic demand analysis.

Forecasting Results and Model Comparison

Two time series forecasting approaches, SARIMA and Holt-Winters Exponential Smoothing, were evaluated to assess their ability to capture temporal demand patterns within the retail dataset.

Model	MAE	RMSE	MAPE
SARIMA	8484.27	12713.24	(%) 36.95
Holt-Winters	7469.57	11863.26	36.89

The results indicate that the Holt-Winters model achieved lower MAE, RMSE, and MAPE values than the SARIMA model, suggesting superior forecasting performance for this demand series.

The relatively similar MAPE values indicate that both models captured the general demand dynamics, although Holt-Winters provided a modest improvement in predictive accuracy. This suggests that the demand series may be characterized more strongly by level, trend, and seasonal components than by the more complex autoregressive structures modeled by SARIMA.

Necessity of Forecasting Demand

Although the subsequent stochastic inventory analysis was based on probabilistic demand modeling rather than forecast outputs, time series forecasting was performed for several important reasons:

- To understand the predictability of historical demand patterns.

- To identify the presence of trend and seasonality in the demand series.
- To evaluate whether future demand can be reasonably estimated from past observations.
- To compare alternative forecasting methodologies commonly used in inventory management.
- To provide additional insight into the temporal behavior of retail demand.

The forecasting exercise demonstrated that demand exhibits measurable temporal structure and can be modeled using statistical time series techniques. These findings complement the stochastic demand analysis by providing a deterministic perspective on demand behavior before uncertainty is explicitly incorporated through probability distributions and Monte Carlo simulation.

Stochastic Demand Modeling

Time-series models provide point forecasts of future demand. However, inventory decisions require more than a single expected value.

Retail demand is inherently uncertain and is influenced by numerous unpredictable factors including customer behaviour, seasonality, promotional activities, and market conditions. Therefore, demand is modeled as a random variable rather than a deterministic quantity.

The objective of this section is to:

1. Quantify demand uncertainty.
2. Identify an appropriate probability distribution.
3. Estimate distribution parameters.
4. Use the fitted distribution for risk analysis and simulation.

This probabilistic framework forms the basis for inventory optimization under uncertainty.

Descriptive Analysis of Demand

Before fitting probability distributions, it is important to examine the basic statistical properties of demand.

Particular attention is paid to:

Mean
Variance
Standard Deviation
Skewness

These measures provide insight into the level of variability present in the demand process.

```
from scipy.stats import skew
print("Mean =", daily_demand["Demand"].mean())
```

```
print("Variance =", daily_demand["Demand"].var())  
  
print("Std Dev =", daily_demand["Demand"].std())  
  
print("Skewness =", skew(daily_demand["Demand"]))
```

```
Mean = 17186.85818181818  
Variance = 108077250.31193098  
Std Dev = 10396.020888394318  
Skewness = 2.5244230647699264
```

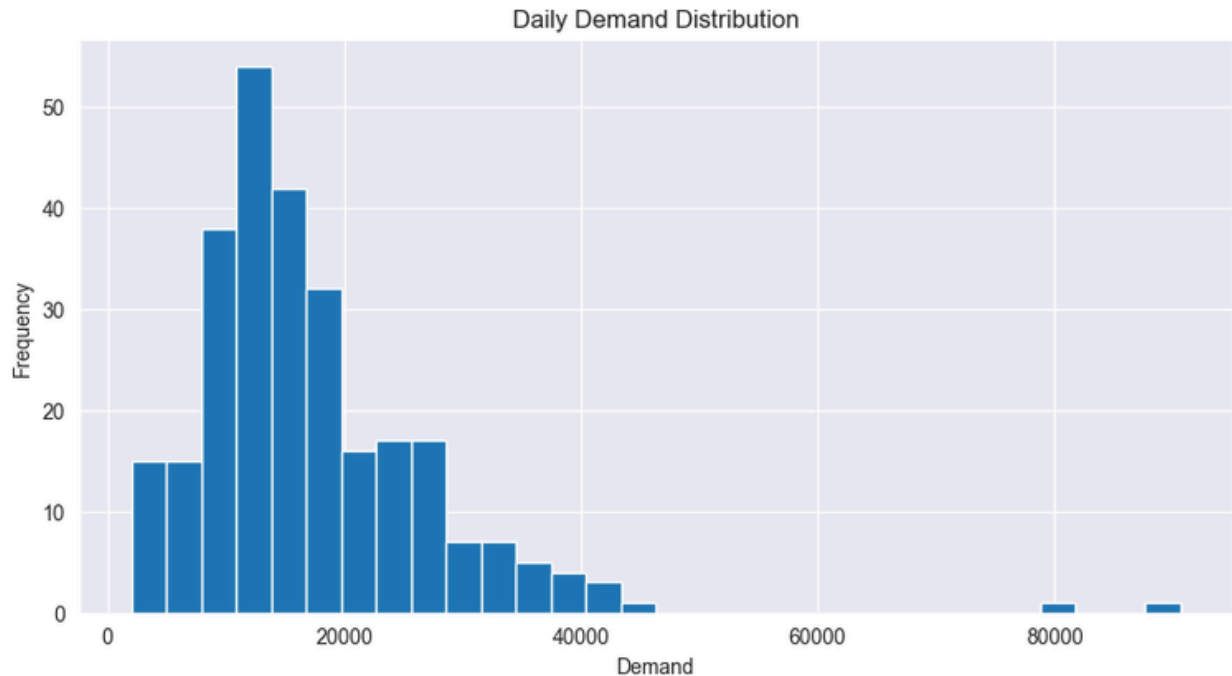
Distribution of Daily Demand

The histogram provides a visual representation of the empirical demand distribution.

A symmetric bell-shaped distribution would indicate relatively stable demand.

Strong right-skewness or heavy tails would suggest the presence of demand spikes and extreme purchasing events.

```
plt.figure(figsize=(10,5)) plt.hist(  
  
    daily_demand["Demand"],  
    bins=30  
)  
plt.title("Daily Demand Distribution")  
plt.xlabel("Demand")  
plt.ylabel("Frequency")  
plt.show()
```



Poisson Demand Model

The Poisson distribution is one of the most widely used stochastic models for count-based demand.

A key property of the Poisson distribution is:

Mean = Variance

Therefore, comparing the sample mean and variance provides an initial assessment of whether Poisson assumptions are reasonable.

```
mean_demand = daily_demand['Demand'].mean()
var_demand = daily_demand['Demand'].var()
print("Mean =", mean_demand)
print("Variance =", var_demand)
print("Variance/Mean Ratio =",
      var_demand/mean_demand)
```

Mean = 17186.85818181818

Variance = 108077250.31193098

Variance/Mean Ratio = 6288.365748328854

Interpretation

For a Poisson process:

Variance $\approx$ Mean

If:

Variance $>$ Mean

the demand process exhibits overdispersion.

Overdispersion indicates that demand variability exceeds what would be expected under a Poisson model and suggests the need for a more flexible distribution.

Negative Binomial Demand Model

Retail demand frequently exhibits overdispersion due to demand spikes, bulk purchases, and heterogeneous customer behaviour.

The Negative Binomial distribution extends the Poisson model by allowing the variance to exceed the mean.

As a result, it often provides a more realistic representation of retail demand uncertainty.

```
mean_demand = daily_demand['Demand'].mean()
var_demand = daily_demand['Demand'].var()
p = mean_demand/var_demand
r = mean_demand*p/(1-p)
print("Estimated r =", r)
print("Estimated p =", p)

Estimated r = 2.7335546983864187
Estimated p = 0.000159023829087192
```

Comparing Empirical and Fitted Distributions

The fitted distribution should resemble the observed demand distribution.

Visual comparison allows us to assess whether the chosen stochastic model adequately captures demand variability.

```
from scipy.stats import nbinom

x = np.arange(
    int(daily_demand['Demand'].min()),
    int(daily_demand['Demand'].max())
```

```

) plt.figure(figsize=(10,5)) plt.hist(

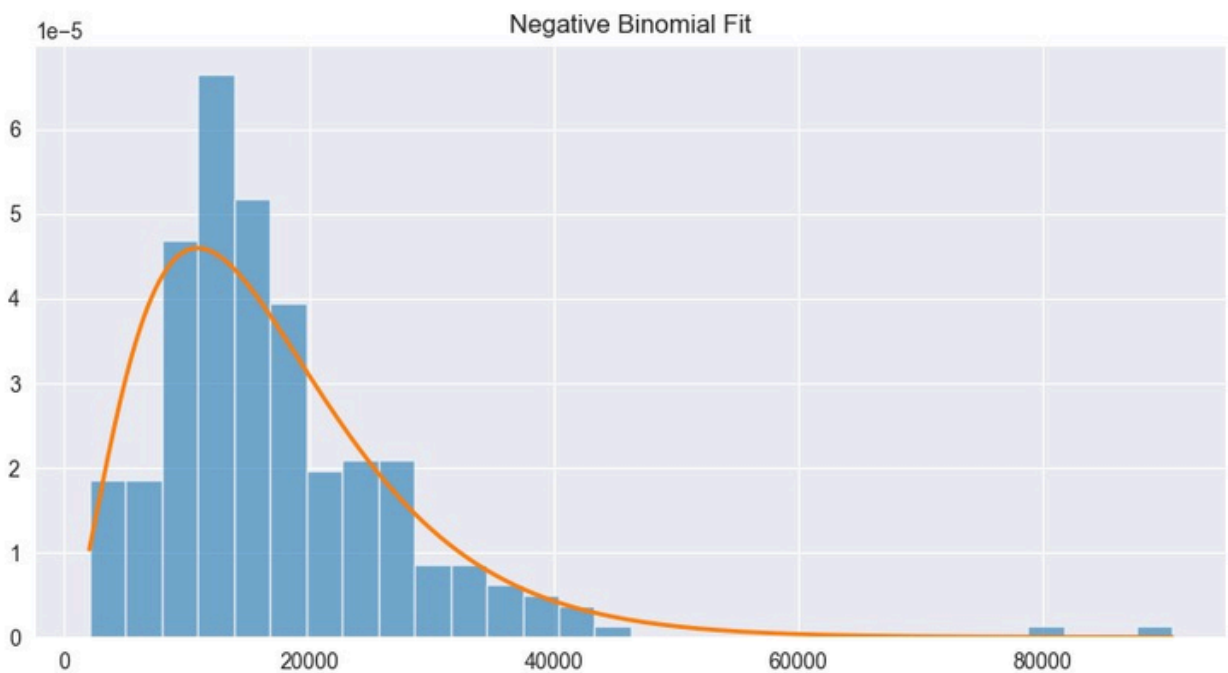
    daily_demand['Demand'],
    bins=30,
    density=True,
    alpha=0.6
)

plt.plot(
    x, nbinom.pmf(x,r,p),
    linewidth=2
)

plt.title(
    "Negative Binomial Fit"
)

plt.show()

```



Fitting Multiple Continuous Probability Distributions

Demand uncertainty plays a critical role in inventory management. Rather than assuming a specific distribution a priori, multiple continuous probability distributions are fitted to the historical demand data and compared using a goodness-of-fit criterion.

The objective is to identify the distribution that best captures the observed demand behavior and can therefore be used for subsequent stochastic inventory calculations.

Candidate Distributions

The following continuous distributions are evaluated:

1. **Normal Distribution**
 - Assumes symmetric demand around the mean.
 - Commonly used in classical inventory models.
2. **Lognormal Distribution**
 - Suitable when demand is strictly positive and right-skewed.
 - Frequently observed in sales and demand data.
3. **Gamma Distribution**
 - Flexible distribution for positive-valued demand.
 - Widely used in reliability and inventory modeling.
4. **Weibull Distribution**
 - Flexible shape characteristics.
 - Can model varying levels of skewness and variability.

Model Selection Using AIC

To determine the most appropriate distribution, the Akaike Information Criterion (AIC) is calculated for each fitted distribution.

$$[AIC = 2k - 2\ln(L)]$$

Where:

- (k) = Number of estimated parameters
- (L) = Maximum likelihood value

The distribution with the lowest AIC value is considered the best balance between model fit and model complexity.

Why Compare Multiple Distributions?

Comparing multiple candidate distributions provides several benefits:

- Reduces the risk of incorrectly assuming a demand distribution.
- Produces more realistic estimates of demand uncertainty. Improves
- safety stock calculations.
- Enhances reorder point estimation.
- Strengthens the robustness of stochastic inventory decisions.

Interpretation

After fitting all candidate distributions, the distribution with the lowest AIC score is selected as the best representation of demand uncertainty.

The selected distribution is then used in subsequent inventory optimization procedures, including:

- Safety Stock Calculation
- Reorder Point Determination
- Service Level Analysis
- Continuous Review (Q,R) Inventory Modeling

This approach ensures that inventory decisions are based on the empirical characteristics of the demand data rather than relying solely on theoretical assumptions.

```
from scipy import stats
import numpy as np
import pandas as pd

demand = daily_demand['Demand']

distributions = {
    "Normal": stats.norm, "Lognormal":
    stats.lognorm, "Gamma": stats.gamma,
    "Weibull": stats.weibull_min
}

results = []

for name, dist in distributions.items():
    params = dist.fit(demand)
    loglikelihood = np.sum(dist.logpdf(demand, *params)) k = len(params)
    aic = 2*k - 2*loglikelihood
    results.append([name, aic])

results = pd.DataFrame(results,

                        columns=["Distribution", 'AIC'])

results.sort_values('AIC')
```

	Distribution	AIC
1	Lognormal	5753.712069
3	Weibull	5771.578509
2	Gamma	5837.475590
0	Normal	5870.462493

Visual Validation

In addition to numerical comparison using AIC, the fitted probability density functions are visually compared against the empirical demand histogram.

A close alignment between the fitted density curve and the observed demand distribution indicates a satisfactory model fit. This graphical assessment complements the quantitative goodness-of-fit evaluation and provides intuitive validation of the selected distribution.

```
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

shape, loc, scale = stats.gamma.fit(demand)

x = np.linspace(
    demand.min(),
    demand.max(),
    1000
)

pdf = stats.gamma.pdf(
    x,
    shape, loc=loc,
    scale=scale
)

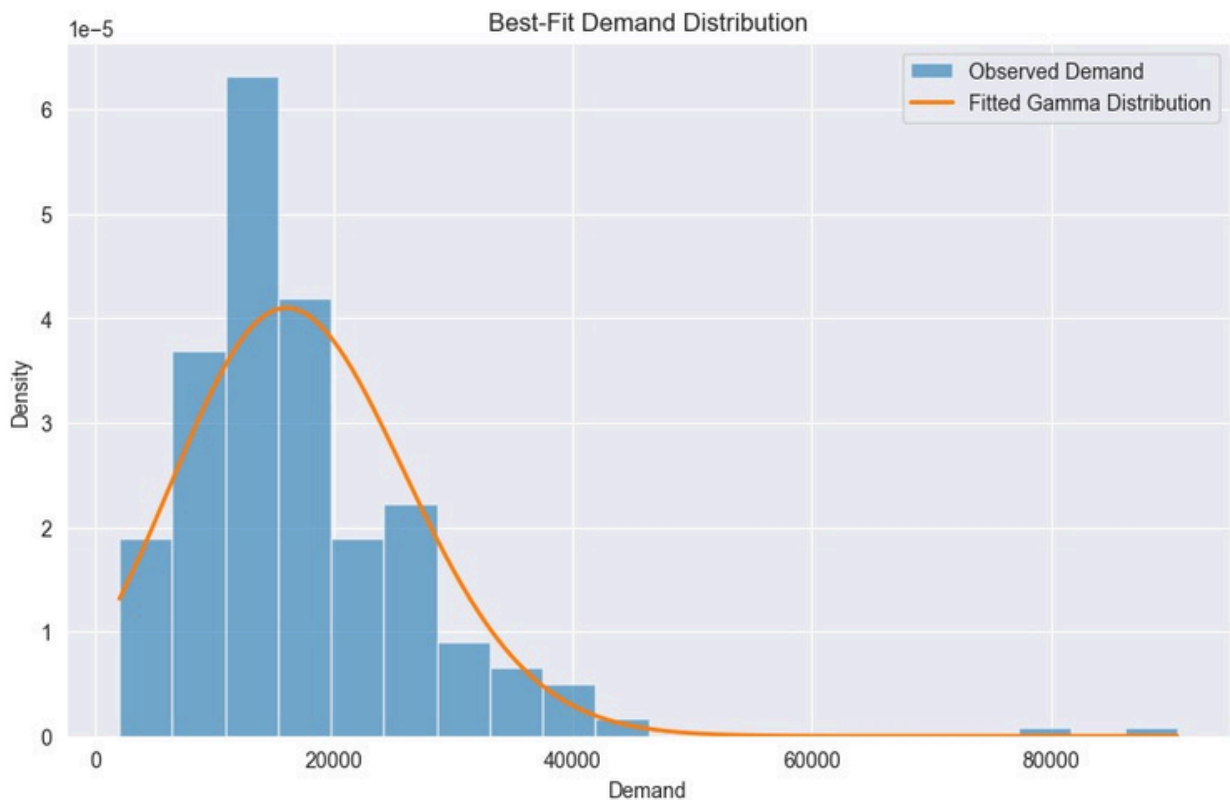
plt.figure(figsize=(10,6))

plt.hist(
    demand,
    bins=20,
    density=True, alpha=0.6,
    label='Observed Demand'
)

plt.plot(
    x,
    pdf,
    linewidth=2,
    label='Fitted Gamma Distribution'
)

plt.title('Best-Fit Demand Distribution')
plt.xlabel('Demand') plt.ylabel('Density')
plt.legend()
```

```
plt.show()
```



Comparative Distribution Fitting for Demand Modeling

Accurately characterizing demand uncertainty is essential for inventory optimization. To identify an appropriate stochastic demand model, multiple candidate distributions were fitted to the historical demand data.

Two distributions were considered:

1. Negative Binomial Distribution

The Negative Binomial distribution is commonly used for count-based demand data and is particularly suitable when demand exhibits overdispersion, i.e., when the variance exceeds the mean.

Advantages:

- Handles count-valued demand naturally.
- Accommodates overdispersion.
- Frequently used in retail and inventory applications.

2. Gamma Distribution

The Gammadistribution is a continuous probability distribution capable of modeling positive and right-skewed demand patterns.

Advantages:

- Flexible shape characteristics.
- Suitable for positively skewed demand.
- Widely used in stochastic inventory theory.

Model Selection

The fitted distributions were evaluated using:

- Akaike Information Criterion (AIC)
- Visual comparison with empirical demand histograms
- Consistency with observed demand characteristics

The distribution exhibiting superior goodness-of-fit was selected as the primary demand model for subsequent Monte Carlo simulation and inventory optimization.

This comparison ensures that inventory decisions are based on the most representative characterization of demand uncertainty.

```
from scipy import stats
from scipy.stats import nbinom
import pandas as pd
import numpy as np

demand = daily_demand['Demand'].values

results = []

# Gamma Distribution
gamma_params = stats.gamma.fit(demand)

gamma_loglik = np.sum(
    stats.gamma.logpdf(
        demand,
        *gamma_params
    )
)

gamma_aic = 2*len(gamma_params) - 2*gamma_loglik

results.append([
    "Gamma",
    gamma_aic
])
```

```
# Negative Binomial Distribution mean =
```

```
np.mean(demand)
var = np.var(demand)
r = mean**2/(var-mean)
p = r/(r+mean)
nb_loglik = np.sum(
    nbinom.logpmf(
        demand,
        r,
        p
    )
)

nb_aic = 2*2 - 2*nb_loglik

results.append([
    "Negative Binomial", nb_aic
])

comparison = pd.DataFrame(
    results,
    columns=[
        "Distribution",
        "AIC"
    ]
)

comparison.sort_values("AIC")
```

	Distribution	AIC
1	Negative Binomial	5763.655019
0	Gamma	5837.475590

```
import numpy as np
import matplotlib.pyplot as plt from scipy
import stats
from scipy.stats import nbinom
```

```
plt.figure(figsize=(10,6))
```

```
# Histogram
```

```
plt.hist(
    demand,
    bins=20,
    density=True, alpha=0.5,
    label="Observed Demand"
```

```

)

# Gamma Fit

x = np.linspace(
    min(demand),
    max(demand),
    1000
)

gamma_pdf = stats.gamma.pdf(
    x,
    *gamma_params
)

plt.plot(
    x,
    gamma_pdf,
    linewidth=2,
    label="Gamma Fit"
)

# Negative Binomial Fit

x_nb = np.arange(
    int(min(demand)),
    int(max(demand))+1
)

nb_pmf = nbinom.pmf(
    x_nb,
    r,
    p
)

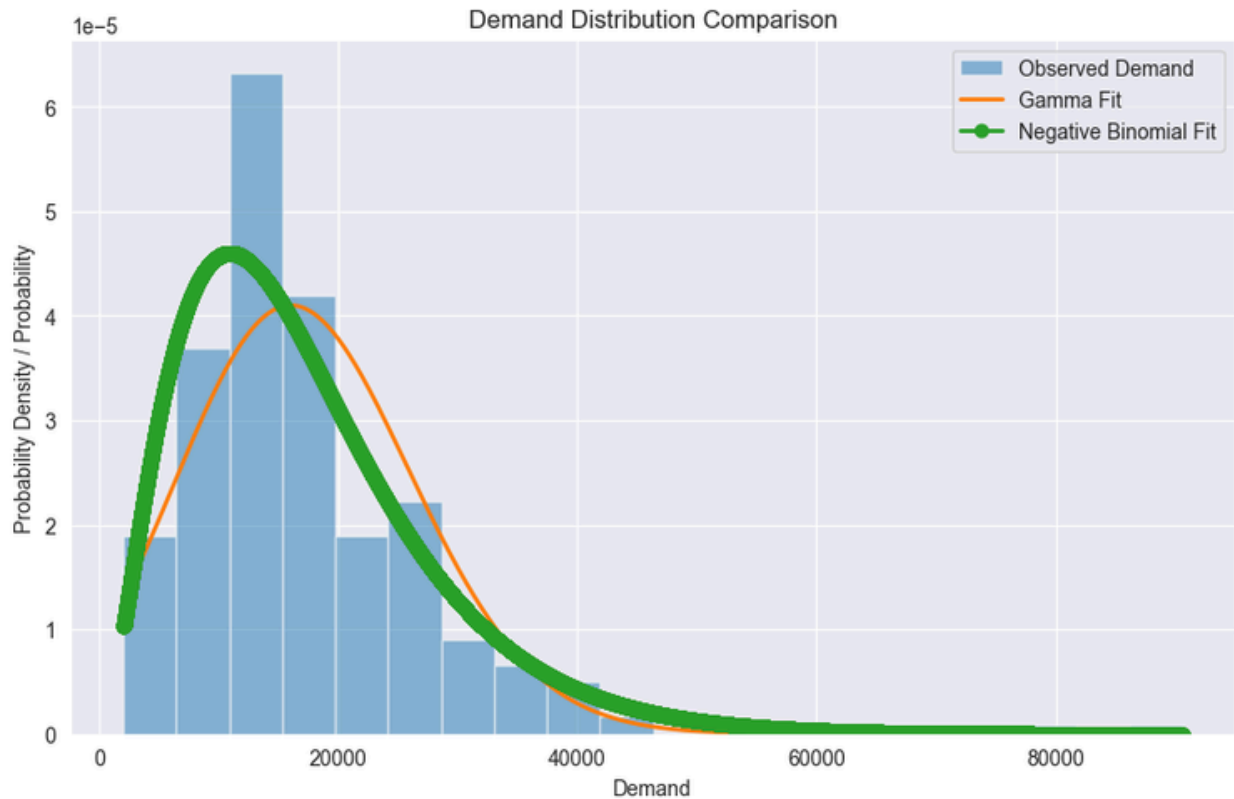
plt.plot(
    x_nb,
    nb_pmf,
    'o-',
    linewidth=2,
    label="Negative Binomial Fit"
)

plt.xlabel("Demand")
plt.ylabel("Probability Density / Probability") plt.title("Demand
Distribution Comparison")

plt.legend()

plt.show()

```



Distribution Selection

Both Gamma and Negative Binomial distributions were fitted to the historical demand data and compared using the Akaike Information Criterion (AIC).

The Negative Binomial distribution achieved a substantially lower AIC than the Gamma distribution, indicating a significantly better fit to the observed demand data.

The superior performance of the Negative Binomial model is consistent with the characteristics of the demand series, which consists of count-valued observations exhibiting overdispersion (variance exceeding the mean).

Consequently, the Negative Binomial distribution was selected as the primary stochastic demand model and was subsequently used to generate demand scenarios through Monte Carlo simulation for inventory policy evaluation.

Monte Carlo Demand Simulation

Demand forecasts provide only a single expected value.

In reality, future demand may follow numerous possible paths.

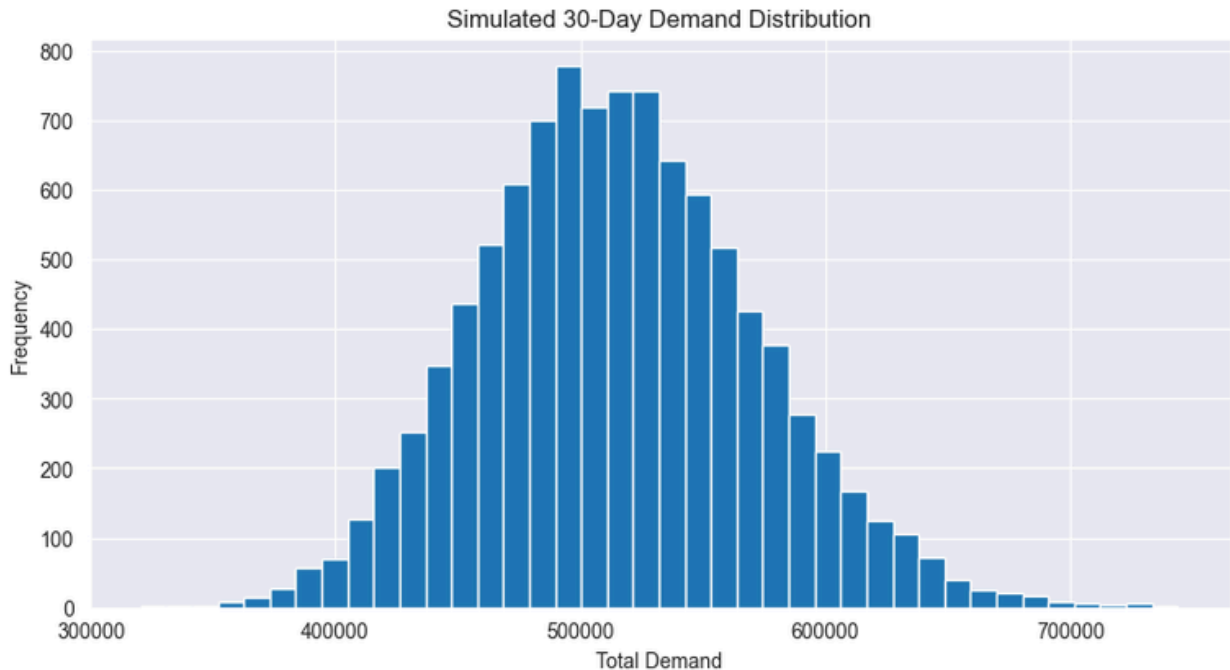
Monte Carlo simulation generates a large number of random demand scenarios using the fitted probability distribution.

This enables the estimation of:

- Inventory risk
- Demand uncertainty
- Stockout probability
- Service performance

under realistic operating conditions.

```
simulations = 10000 future_days = 30
simulated_demand = nbinom.rvs(
    r,
    p, size=(simulations,future_days)
)
total_future_demand = simulated_demand.sum(axis=1)
plt.figure(figsize=(10,5))
plt.hist(
    total_future_demand,
    bins=40
)
plt.title(
    "Simulated 30-Day Demand Distribution"
)
plt.xlabel(
    "Total Demand"
)
plt.ylabel(
    "Frequency"
)
plt.show()
```



Summary of Simulated Demand

The simulated demand distribution provides an estimate of the range of future demand outcomes.

Unlike a single forecast, simulation quantifies both expected demand and uncertainty around that expectation.

```
print(  
    "Mean Simulated Demand =",  
    total_future_demand.mean()  
)  
  
print(  
    "Std Dev =",  
    total_future_demand.std()  
)  
  
print(  
    "95th Percentile =",  
    np.percentile(  
        total_future_demand, 95  
    )  
)
```

```
Mean Simulated Demand = 514985.5403  
Std Dev = 56304.39516004746  
95th Percentile = 611646.7999999999
```

Stockout Probability Analysis

A stockout occurs when available inventory is insufficient to satisfy customer demand.

Using simulated demand scenarios, the probability of stockout can be estimated directly. This provides a quantitative measure of inventory risk.

```
inventory_level = 500000

stockout_probability = np.mean(
    total_future_demand >
    inventory_level
)

print(
    "Stockout Probability =",
    stockout_probability
)
```

```
Stockout Probability = 0.8472
```

Inventory Risk Assessment and Need for Optimization

The Monte Carlo simulation revealed a stockout probability of approximately 84%.

This result indicates that under the current inventory policy, there is a 84% chance that available inventory will be insufficient to satisfy future customer demand during the planning horizon.

Such a high probability of stockout is undesirable for several reasons:

- Loss of potential sales revenue.
- Reduced customer satisfaction.
- Increased risk of customer attrition.
- Disruptions in supply chain performance.
- Reduced service levels.

The analysis demonstrates that relying solely on historical averages or simple forecasting models is insufficient for effective inventory management.

Although machine learning models provided useful demand forecasts, substantial uncertainty remained in the demand process. This uncertainty was further confirmed through stochastic demand modeling and Monte Carlo simulation.

Consequently, a more systematic inventory policy is required.

The objective is not merely to forecast demand but to determine inventory decisions that balance:

- Stockout risk
- Ordering costs

- Holding costs
- Service level requirements

To address these challenges, Operations Research techniques are now employed to develop optimized inventory policies.

Specifically, the following questions will be addressed:

1. What inventory level is required to achieve a desired service level?
2. How much safety stock should be maintained?
3. At what inventory level should replenishment be triggered?
4. What order quantity minimizes total inventory cost?

The next section applies inventory optimization models to transform demand forecasts and stochastic demand estimates into actionable inventory management decisions.

Service Level Analysis

Inventory managers must determine the desired probability of satisfying customer demand without experiencing a stockout.

This probability is known as the Service Level.

A higher service level reduces stockout risk but requires additional inventory investment.

Common industry benchmarks include:

- 90% Service Level
- 95% Service Level
- 99% Service Level

The objective of this section is to quantify the inventory implications of different service level targets.

Service Level and Z-Scores

Under the assumption of approximately normal demand during lead time, safety stock can be calculated using standard normal quantiles.

Higher service levels correspond to larger Z-values and therefore greater safety stock requirements.

service_levels = {

```
'90%':1.28,
'95%':1.65,
'99%':2.33
}
```

service_levels

```
{'90%': 1.28, '95%': 1.65, '99%': 2.33}
```

Safety Stock Calculation

Safety stock is additional inventory maintained to protect against demand uncertainty during lead time.

The safety stock formula is:

$$\text{Safety Stock} = z \times \sigma \times \sqrt{L}$$

where:

- z = service level factor
- σ = demand standard deviation
- L = lead time

Safety stock acts as a buffer against unexpected demand fluctuations.

```
sigma = daily_demand['Demand'].std()
```

```
L = 5
```

```
for level, z in service_levels.items():
```

```
    ss = z*sigma*np.sqrt(L)
```

```
    print(level,
          "Safety Stock =",
          round(ss,2))
```

```
90% Safety Stock = 29755.15 95%
Safety Stock = 38356.25 99% Safety
Stock = 54163.67
```

Reorder Point Analysis

The Reorder Point (ROP) specifies the inventory level at which a replenishment order should be placed.

The objective is to ensure that inventory remains sufficient throughout the lead time period.

ROP incorporates:

1. Expected demand during lead time.
2. Safety stock.

This allows the inventory system to account for both average demand and uncertainty.

```
d = daily_demand['Demand'].mean()
```

```
rop_results = []
```

```
for level, z in service_levels.items():
```

```

ss = z*sigma*np.sqrt(L) rop =
d*L + ss rop_results.append([

    level,
    round(ss,2),
    round(rop,2)
])
rop_df = pd.DataFrame(
    rop_results,
    columns=[
        'Service Level', 'Safety
        Stock', 'ROP'
    ]
)
rop_df

```

	Service Level	Safety Stock	ROP
0	90%	29755.15	115689.44
1	95%	38356.25	124290.54
2	99%	54163.67	140097.96

Interpretation of Reorder Point Results

As service levels increase, both safety stock and reorder points increase.

This reflects the fundamental trade-off in inventory management:

- Higher service levels reduce stockout risk.
- Higher service levels increase inventory holding costs.

Organizations must therefore balance customer service objectives against inventory investment requirements.

Economic Order Quantity (EOQ)

While the reorder point determines when to place an order, the Economic Order Quantity determines how much inventory should be ordered.

EOQ minimizes the total inventory cost by balancing:

- Ordering costs
- Holding costs

The EOQ model is one of the most widely used inventory optimization techniques in operations management.

```
annual_demand = daily_demand['Demand'].sum() S = 500
```

```
H = 20
```

```
EOQ = np.sqrt(
```

```
    (2*annual_demand*S)/H
)
```

```
print("EOQ =", round(EOQ,2))
```

```
EOQ = 15372.68
```

Interpretation of EOQ

The EOQ represents the order quantity that minimizes total inventory cost.

Ordering quantities smaller than EOQ may increase ordering frequency and associated ordering costs.

Ordering quantities larger than EOQ may increase holding costs due to excessive inventory accumulation.

Therefore, EOQ provides a cost-efficient replenishment quantity under the assumptions of the model.

Inventory Cost Analysis

Inventory decisions involve two primary cost components:

1. Ordering Cost
2. Holding Cost

The EOQ model identifies the point at which these costs are balanced.

The following analysis estimates the annual cost associated with the optimized inventory policy.

```
ordering_cost = (
    annual_demand/EOQ
)*S
```

```
holding_cost = (
    EOQ/2
)*H
```

```
total_cost = (
    ordering_cost
    +
    holding_cost
)
```

```
cost_df = pd.DataFrame({
    'Cost Component': [
        'Ordering Cost',
        'Holding Cost', 'Total
        Cost'
    ], 'Amount': [
        ordering_cost,
        holding_cost,
        total_cost
    ]
})
```

cost_df

	Cost Component	Amount
0	Ordering Cost	153726.803128
1	Holding Cost	153726.803128
2	Total Cost	307453.606256

Interpretation of Cost Results

The inventory cost analysis provides insight into the financial implications of the proposed inventory policy.

The ordering cost represents the expenditure associated with placing and processing replenishment orders, while the holding cost reflects the expenses incurred from storing inventory over time.

The total inventory cost combines these two components and represents the annual cost of operating the inventory system under the EOQ framework.

The results demonstrate the fundamental trade-off in inventory management. Frequent ordering reduces holding costs but increases ordering expenses, whereas large order quantities reduce ordering frequency but increase inventory carrying costs.

The EOQ obtained in this study balances these competing cost components and provides a cost-efficient replenishment policy.

When combined with the previously determined safety stock and reorder point levels, the proposed inventory strategy not only minimizes costs but also helps reduce stockout risk and improve service performance under uncertain demand conditions.

Conclusion

This study developed an integrated inventory management framework combining Time Series Forecasting, Stochastic Modeling, Monte Carlo Simulation, and Operations Research techniques using the UCI Online Retail dataset.

The analysis began with data preprocessing, demand aggregation, and exploratory analysis to understand historical demand patterns. To assess demand predictability, two statistical time series forecasting approaches, SARIMA and Holt-Winters Exponential Smoothing, were implemented and evaluated. Forecasting performance was assessed using MAE, RMSE, and MAPE metrics. Among the models considered, Holt-Winters achieved slightly superior forecasting accuracy, indicating that trend and seasonal components played an important role in explaining demand behavior. Nevertheless, both models exhibited substantial forecasting errors, highlighting the inherent uncertainty present in retail demand.

To explicitly account for this uncertainty, demand was subsequently modeled using probabilistic methods. A comparative analysis of Gamma and Negative Binomial distributions was conducted, and the Negative Binomial distribution was selected as the preferred stochastic demand model due to its substantially superior goodness-of-fit. This result suggested that the demand process exhibited characteristics consistent with overdispersed count data.

Monte Carlo simulation was then employed to generate a large number of plausible future demand scenarios. The simulation results demonstrated the extent of demand variability and provided a quantitative assessment of inventory risk under uncertain operating conditions. The analysis highlighted the limitations of relying solely on deterministic demand estimates when making inventory decisions.

To address inventory risk, Operations Research techniques were applied to determine safety stock levels, reorder points, and economically optimal order quantities. A Continuous Review (Q,R) inventory framework was utilized to develop inventory policies capable of balancing service-level requirements against inventory-related costs.

Overall, the study demonstrates that effective inventory management requires not only an understanding of historical demand patterns but also explicit consideration of uncertainty and risk. By integrating forecasting, probabilistic modeling, simulation, and optimization techniques, the proposed framework provides a comprehensive and practical approach for inventory planning in environments characterized by uncertain demand.

The findings emphasize the value of combining statistical analysis, stochastic modeling, and Operations Research methodologies to support more robust inventory decisions, improve service levels, and reduce the overall cost of inventory management.

Future Scope

The framework developed in this study can be extended in several directions to improve forecasting accuracy, enhance inventory decision-making, and better reflect real-world operational complexities.

1. Incorporation of external factors such as holidays, promotional campaigns, economic conditions, and special events that may influence customer demand and purchasing behavior.
2. Exploration of more advanced forecasting approaches, including machine learning and deep learning techniques such as Long Short-Term Memory (LSTM) networks, Temporal Fusion Transformers (TFT), and hybrid forecasting models that combine statistical and data-driven methods.

3. Development of product-level forecasting and inventory optimization models to capture differences in demand patterns across individual products rather than relying on aggregate demand analysis.
4. Extension to multi-item inventory systems where multiple products compete for limited storage capacity, budget, or operational resources.
5. Implementation of dynamic and adaptive inventory policies that update reorder points and safety stock levels in response to evolving demand patterns and changing business conditions.
6. Incorporation of supplier lead-time uncertainty and supply-side disruptions into the stochastic modeling framework to provide a more comprehensive assessment of inventory risk.
7. Evaluation of alternative stochastic demand models and simulation techniques to further improve the representation of demand variability and uncertainty.
8. Integration of service-level optimization and stockout cost analysis to support inventory policies that explicitly balance customer satisfaction with operational costs.
9. Development of real-time inventory monitoring and decision-support systems capable of continuously updating forecasts, risk estimates, and replenishment recommendations as new data become available.

These extensions would further strengthen the practical applicability of the framework and enable more realistic, responsive, and data-driven inventory management systems for modern retail and supply chain environments.

Summary

This project developed an integrated framework combining Time Series Analysis, Stochastic Modeling, Monte Carlo Simulation, and Operations Research techniques for inventory management using the UCI Online Retail dataset.

The study began with data preprocessing, demand aggregation, feature construction, and exploratory data analysis to understand sales patterns and demand behavior over time. Historical retail transactions were transformed into a daily demand time series, enabling the investigation of trends, seasonality, and temporal variability in customer demand.

To assess the predictability of demand, two statistical time series forecasting approaches, SARIMA and Holt-Winters Exponential Smoothing, were implemented and evaluated. Forecast accuracy was measured using MAE, RMSE, and MAPE. The results indicated that the Holt-Winters model marginally outperformed SARIMA, suggesting that the demand series was influenced primarily by trend and seasonal components. However, both models exhibited substantial forecasting errors, highlighting the inherent uncertainty and variability present in retail demand.

Recognizing that demand could not be fully characterized through deterministic forecasting alone, the study proceeded to stochastic demand modeling. Multiple probability distributions were evaluated to represent demand uncertainty. A comparative analysis between the Gamma and Negative Binomial distributions was conducted using goodness-of-fit measures. The Negative Binomial distribution achieved a substantially lower AIC value and was therefore selected as the primary stochastic demand model, indicating that the demand data exhibited characteristics consistent with overdispersed count processes.

Monte Carlo simulation was subsequently performed using the fitted Negative Binomial distribution to generate a large number of plausible future demand scenarios. This simulation framework enabled the quantification of demand uncertainty and provided insights into inventory risk under varying demand conditions.

To translate these insights into actionable inventory policies, Operations Research techniques were applied. Safety stock requirements were estimated to protect against demand variability, reorder points were calculated to support timely replenishment decisions, and the Economic Order Quantity (EOQ) model was utilized to determine cost-efficient order quantities. A Continuous Review (Q,R) inventory policy was further examined to balance service levels with inventory-related costs.

Inventory cost analysis highlighted the trade-offs between ordering costs, holding costs, and stockout risk, demonstrating the importance of incorporating uncertainty into inventory planning decisions. The results showed that inventory policies based solely on average demand may underestimate risk and lead to insufficient inventory levels under uncertain operating conditions.

Overall, the study demonstrates that effective inventory management requires not only an understanding of historical demand patterns but also explicit consideration of uncertainty and risk. By integrating time series forecasting, probabilistic demand modeling, Monte Carlo simulation, and inventory optimization techniques, the proposed framework provides a comprehensive and data-driven approach to inventory planning in environments characterized by uncertain demand.

The findings highlight the value of combining forecasting, stochastic analysis, and optimization methodologies to support more robust inventory decisions, improve service levels, and reduce the overall cost of inventory management.

Refereneces

UCI Online Retail Dataset
Pandas Documentation
Scikit-Learn Documentation
SciPy Documentation
Silver, Pyke & Peterson - Inventory Management and Production Planning
Montgomery et al. - Introduction to Statistical Quality Control